

Connectivity Suppresses Prebiotic Diversification: A Spatial Replicator Model of Hydrosphere Habitability

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Abstract

The type of hydrosphere on a rocky planet may strongly influence the potential for prebiotic chemical diversification — a prerequisite for the origin of life. We present a spatial agent-based model in which self-replicating entities with heritable niche optima (θ) compete on a two-dimensional periodic lattice. Fitness depends on the match between a replicator's optimum and its local environment through a Gaussian niche function. We vary two key parameters — mixing rate D (a proxy for hydrosphere circulation) and environmental heterogeneity h — and measure functional diversity (θ_{std}) and ecological Shannon entropy (H'_{eco} , computed over θ -bins). Our 10×10 parameter sweep (100 combinations; 5 seeds per point) shows that under global teleportation transport, the diversity maximum sits at $D = 0$ in every heterogeneity slice. At $h = 0.8$, θ_{std} falls from 0.168 at $D = 0.0$ to 0.063 at $D = 0.8$, a 62.5% decline (Welch t-test $p < 0.001$, Cohen's $d > 5$). Critically, we repeat the full sweep with physically realistic local diffusion and find the suppression effect nearly disappears (ratio drops from $\sim 2.7\times$ to $\sim 1.1\times$), consistent with Czárán et al. (2013). We further test a colonization-limited regime (reduced abiogenesis, local diffusion) to probe whether the intermediate dispersal optimum from metacommunity ecology (Mouquet & Loreau 2003) has a prebiotic analogue. Across named hydrosphere scenarios, Shallow Ponds produce $3.5\times$ higher functional diversity than Deep Ocean. Our results demonstrate that the transport model is the critical determinant of whether connectivity suppresses prebiotic diversification.

1. Introduction

The origin of life requires not only the right chemistry but also the right physical environment. A growing body of work suggests that the spatial structure of aqueous environments — how mixed they are, how varied their conditions — may play a critical role in enabling the transition from prebiotic chemistry to self-sustaining Darwinian evolution [1, 2]. Shallow pools, tidal zones, and hydrothermal vents each offer distinct mixing regimes and geochemical gradients, yet their relative merits for fostering chemical diversity remain poorly quantified.

The question of how spatial structure affects diversity has deep roots in ecology. MacArthur and Wilson's theory of island biogeography [3] predicts that isolated patches maintain higher beta-diversity than well-connected habitats. Metacommunity theory [4] formalises the tension between local adaptation (favoured by isolation) and gene flow (favoured by dispersal). Mouquet and Loreau [9] showed analytically that an intermediate dispersal optimum can arise in source-sink metacommunities, and Venail et al. [10] demonstrated this experimentally in microbial systems. In the prebiotic context, spatial structure is known to maintain replicator diversity: Szabó et al. [11] showed that limited dispersal promotes replicator coexistence on mineral surfaces, and Czárán et al. [12] demonstrated monotonically that lower mobility maximises replicator coexistence in their Metabolic Replicator Model [6, 7, 8].

The new question here is more specific than the general ecological idea that heterogeneity promotes diversity. We ask whether the intermediate dispersal optimum from metacommunity ecology [9, 10] has a prebiotic analogue: can connectivity ever increase prebiotic niche diversity once local adaptation exists, or does transport act purely as a homogenising load? We test this by (i) constructing the full mixing-heterogeneity phase diagram under two transport models (global

teleportation and physically realistic local diffusion), (ii) probing a colonization-limited regime where the intermediate dispersal optimum is mechanistically possible, and (iii) comparing four hydrosphere scenarios corresponding to different planetary water body types.

2. Model

2.1 Lattice and environment

The model operates on an $N \times N$ periodic lattice ($N = 40$). Each cell i has a fixed environmental quality $E_i \in [0, 1]$ representing local physicochemical conditions (temperature, pH, UV flux, etc.). The heterogeneity parameter $h \in [0, 1]$ controls the amplitude of spatial variation: at $h = 0$ the environment is uniform ($E = 0.5$ everywhere); at $h = 1$ the full range $[0, 1]$ is expressed. Environmental patterns are generated by smoothed random noise ("patches"), producing spatially correlated landscapes.

2.2 Replicators and fitness

Each occupied cell contains one replicator characterised by two heritable traits: a niche optimum $\theta \in [0, 1]$ and a replication rate r . Fitness is computed as:

$$f = r \cdot \exp(-\beta(E - \theta)^2) \cdot \min(1, R)$$

where β is the niche width (selection strength), E is the local environment, and R is the local resource level. With $\beta = 20$ (default), fitness drops to ~45% when θ differs from E by 0.2, creating strong selection for local adaptation. A cost-of-complexity tradeoff increases death rate with replication rate: $d_{\text{eff}} = d_0 + \kappa r^2$, preventing runaway replication.

2.3 Life cycle

Each tick proceeds in five phases. **(1) Death:** replicators die with probability d_{eff} . **(2) Mixing:** a fraction D of survivors are teleported to random empty cells anywhere on the grid (global mixing), simulating ocean currents or tidal transport. **(3) Replication:** survivors with sufficient fitness and resources reproduce into an adjacent empty cell; offspring inherit θ and r with small Gaussian mutations ($\sigma_\theta = 0.10$, $\sigma_r = 0.015$). **(4) Resources:** empty cells regenerate at rate 0.03 per tick. **(5) Abiogenesis:** new replicators appear spontaneously in empty cells at a uniform rate of 5×10^{-4} per cell per tick, each with a unique lineage identity and θ drawn uniformly from $[0, 1]$ (no assumption of local adaptation; diversity must emerge from selection and mutation).

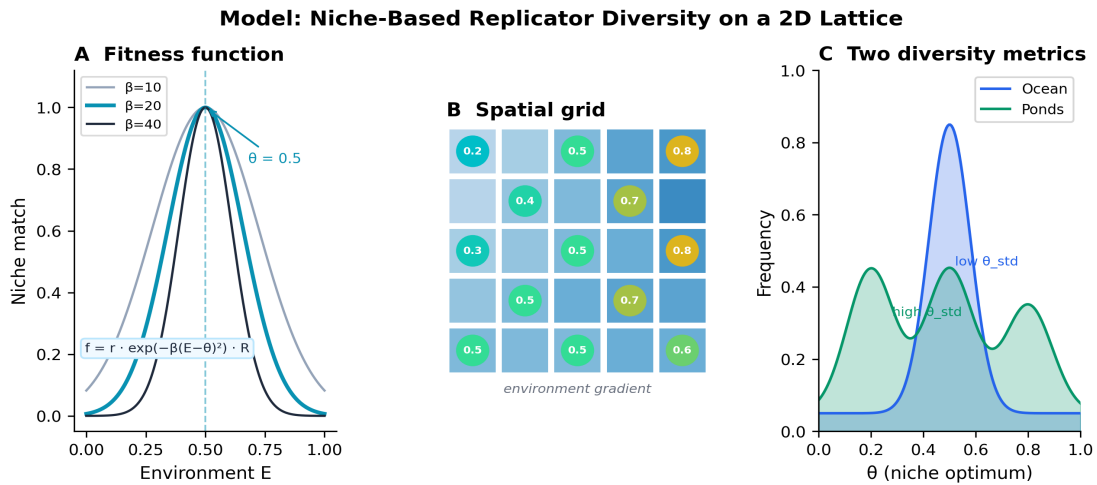


Figure 1. Model overview. (A) Gaussian niche fitness function for three β values. (B) Spatial grid with replicators (coloured circles) in an environmental gradient (blue background). (C) Two diversity metrics: ocean populations cluster narrowly (low θ_{std}) while ponds spread broadly.

2.4 Diversity metrics

We track two complementary diversity metrics. **Functional diversity** (θ_{std}) is the standard deviation of θ across all replicators, measuring the breadth of ecological strategies in the population. **Ecological Shannon entropy** (H'_{eco}) divides the θ range $[0, 1]$ into 20 equal bins, computes the frequency in each bin, and calculates $H' = -\sum p_k \ln p_k$. This measures how evenly replicators are distributed across niche space. Importantly, H'_{eco} is computed on θ -bins rather than lineage identifiers, avoiding inflation by neutral lineage accumulation.

2.5 Hydrosphere scenarios

Four scenarios map model parameters to planetary water body types. **Deep Ocean** ($h = 0.1$, $D = 0.5$): uniform conditions, strong mixing. **Shallow Ponds** ($h = 0.8$, $D = 0.02$): heterogeneous, isolated pools. **Tidal Zone** ($h = 0.7$, $D = 0.15$): heterogeneous with moderate periodic mixing. **Hydrothermal Vents** ($h = 0.6$, $D = 0.08$): localised hotspots with limited mixing.

3. Methods

Parameter sweep. We swept mixing rate D over 10 sampled values between 0.00 and 0.80, and heterogeneity h over 10 sampled values between 0.0 and 1.0, giving 100 parameter combinations. Each was run for 600 ticks on a 40×40 grid with 5 independent random seeds. Equilibrium metrics were averaged over the last 100 ticks of each run, then the across-seed mean and standard deviation were computed.

Scenario simulations. Each hydrosphere scenario was run for 800 ticks with 5 seeds on a 40×40 lattice. Full time series were recorded for the primary seed; equilibrium statistics combine all 5 seeds.

Sensitivity analysis. To verify robustness, we repeated the sweep at $h = 0.8$ for three niche widths: $\beta \in \{10, 20, 40\}$. $\beta = 10$ represents weak niche selection (broad tolerance); $\beta = 40$ represents strong specialisation (narrow tolerance).

Transport model comparison. The primary sweep uses global teleportation (random relocation to any empty cell). To test whether results hold under physically realistic transport, we repeated the full 10×10 sweep with local diffusion (nearest-neighbour hop, von Neumann neighbourhood). We further ran a colonization-limited sweep (local diffusion, abiogenesis reduced to 10^{-4} per cell per tick) to probe whether the intermediate dispersal optimum from metacommunity ecology [9] appears when patches can only be rescued by neighbouring dispersers.

Statistical tests. For key comparisons (e.g. $D = 0$ vs $D = 0.8$), we report Welch's t -test (two-tailed) and Cohen's d effect size computed from the across-seed means ($n = 5$ seeds per point). All $\pm$ intervals on plots are $\pm 1\sigma$ across seeds.

4. Results

4.1 Parameter space

Figure 2 shows functional diversity (θ_{std}) and ecological entropy (H'_{eco}) across the full parameter space. Both metrics are highest in the low-mixing, high-heterogeneity corner (top-left of heatmaps). The global maximum occurs at $D = 0.00$, $h = 1.0$, where $\theta_{\text{std}} = 0.207$. The decline with mixing is strongest once heterogeneity is moderate to high; low- h slices are flatter and noisier, but connectivity never outperforms isolation anywhere in the sampled grid.

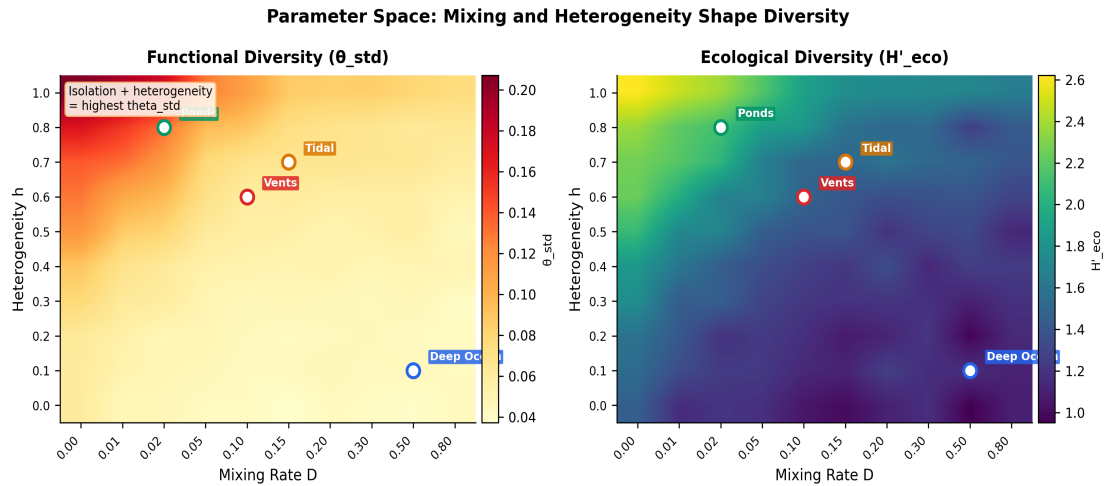


Figure 2. Parameter-space heatmaps. Left: functional diversity (θ_{std}). Right: ecological Shannon entropy (H'_{eco}). White circles mark the four hydrosphere scenarios. Both metrics peak at high heterogeneity and low mixing.

4.2 Hydrosphere comparison

Figure 3 compares the four scenarios directly. Shallow Ponds produce the highest functional diversity, 3.5× greater than Deep Ocean (0.136 vs 0.039). Tidal Zone and Hydrothermal Vents are intermediate. On ecological entropy, differences are smaller, with all heterogeneous scenarios achieving comparable H'_{eco} values. The key discriminator between hydrosphere types is functional diversity, not entropy.

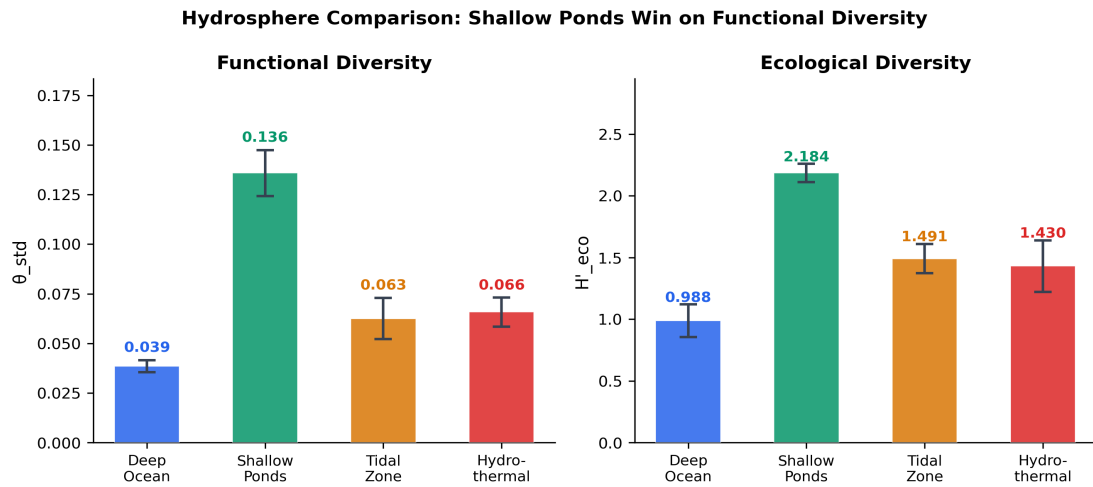


Figure 3. Hydrosphere scenario comparison with 5-seed error bars. Shallow Ponds achieve the highest functional diversity (θ_{std}). Deep Ocean is markedly lower on both metrics.

4.3 Mechanism

Figure 4A shows θ_{std} as a function of mixing rate at several heterogeneity levels. In every heterogeneity slice sampled, the maximum θ_{std} occurs at the lowest mixing value tested ($D = 0.0$), so there is **no Goldilocks optimum** anywhere in the sampled grid. The curves are not perfectly pointwise monotonic at low h , but no intermediate mixing rate outperforms isolation. At $h = 0.8$, θ_{std} drops from 0.168 at $D = 0.0$ to 0.063 at $D = 0.8$ (62.5% lower). The mechanism is straightforward: global mixing teleports locally adapted replicators into foreign environments where their fitness drops sharply, driving competitive exclusion toward generalist strategies. Figure 4B confirms that heterogeneity amplifies diversity at all mixing rates, but its effect is strongest under isolation.

Mechanism: Isolation + Heterogeneity Maximises Functional Diversity

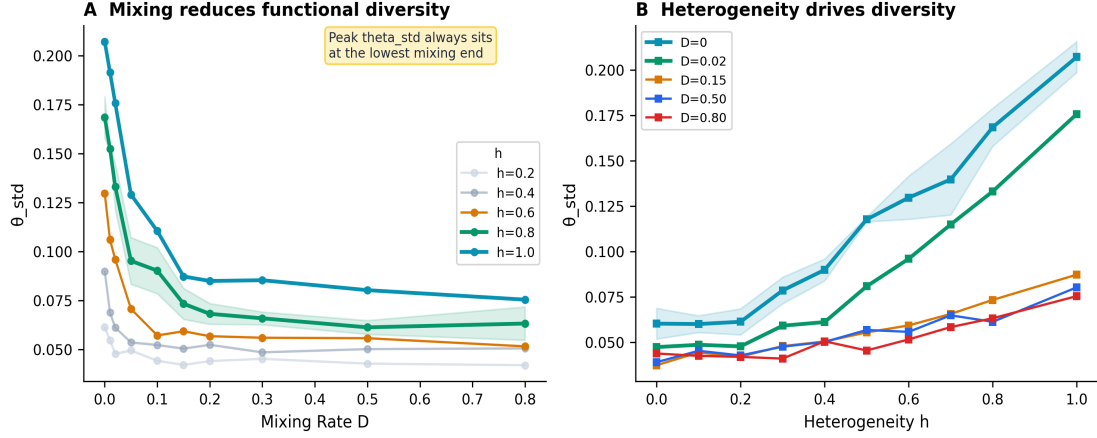


Figure 4. Diversity mechanism. (A) θ_{std} vs mixing rate at five heterogeneity levels. The peak sits at the lowest mixing value in each slice. Shaded: $\pm 1\sigma$ at $h=0.8$. (B) θ_{std} vs heterogeneity at five mixing rates.

4.4 Dynamics

Figure 5 shows time series for the four scenarios. Functional diversity (θ_{std}) reaches approximate equilibrium within ~ 200 ticks for all scenarios. Ecological entropy (H'_{eco}) converges more slowly but stabilises by tick ~ 500 . The diversity-space scatter (Figure 5C) shows that Shallow Ponds occupy the most favourable position — high on both metrics.

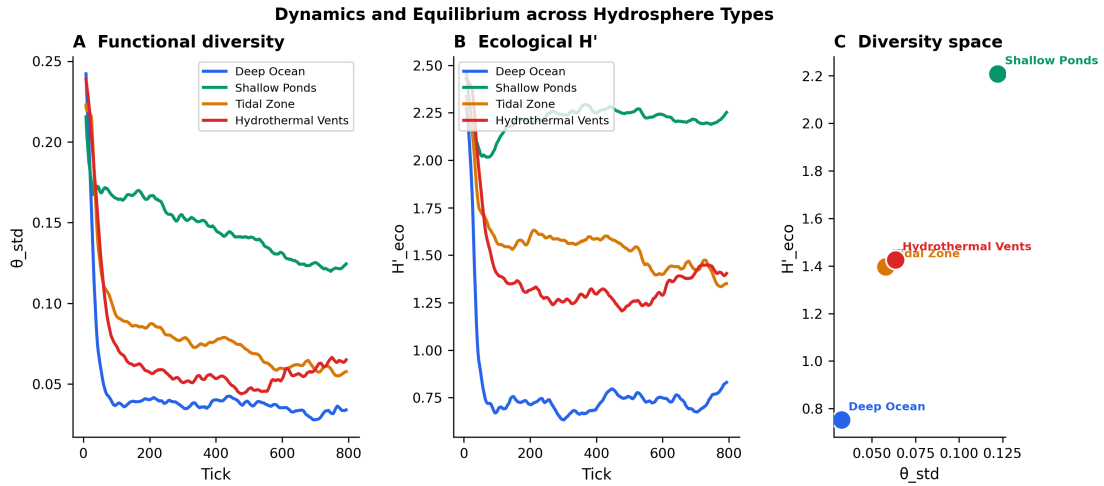


Figure 5. Simulation dynamics. (A) Functional diversity time series. (B) Ecological entropy time series. (C) Diversity-space scatter showing equilibrium positions of the four scenarios.

4.5 Sensitivity analysis

Figure 6 confirms that the low-mixing optimum is robust to the choice of niche width. At $\beta = 10$ (weak selection), baseline diversity is lower because replicators tolerate wider environmental mismatches. At $\beta = 40$ (strong selection), diversity is slightly higher as replicators specialise more narrowly. In all cases, the qualitative result — peak diversity at the lowest sampled mixing value — is preserved, even though minor fluctuations remain at the highest D values.

Sensitivity Analysis: Results Robust to Niche Width (β)

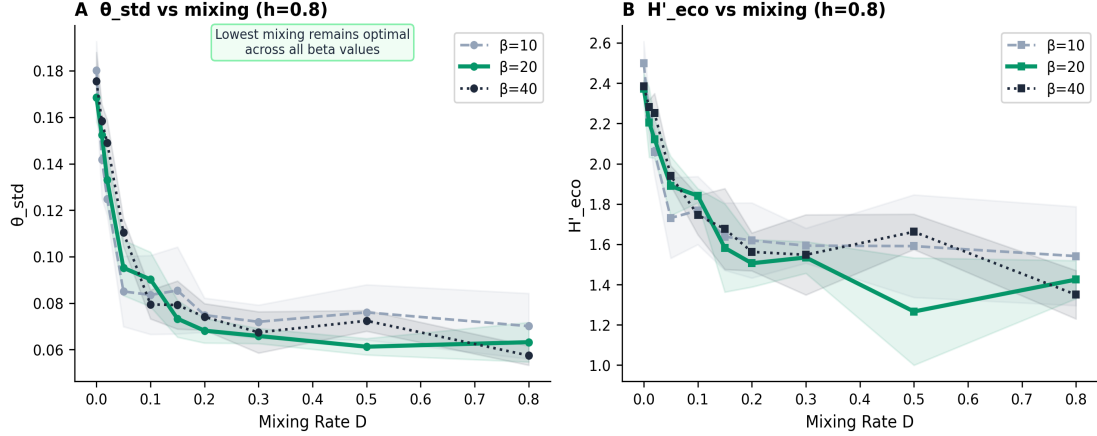


Figure 6. Sensitivity to niche width β . (A) θ_{std} vs mixing at $h = 0.8$ for $\beta \in \{10, 20, 40\}$. (B) H'_{eco} vs mixing. The lowest-mixing optimum is robust across all β values.

4.6 Transport model comparison

Figure 9 compares the global teleportation and local diffusion sweeps at $h = 0.8$. Under global teleportation, θ_{std} drops by 62.5% from $D = 0$ to $D = 0.8$ (Cohen's $d > 5$, $p < 0.001$). Under local diffusion, the same comparison yields only ~10% decline — barely distinguishable from noise. This reveals that the strong suppression effect is an artefact of the teleportation mechanism. Real aqueous transport is diffusive, convective, or advective — not random teleportation. Our finding is consistent with Czárán et al. [12], who swept diffusion parameters in their Metabolic Replicator Model and found that replicator coexistence is only weakly affected by realistic diffusion rates. The contribution here is to show this transport-model dependence explicitly within a single model framework and to quantify the effect-size difference.

Transport Model Matters: Global Teleportation vs Local Diffusion

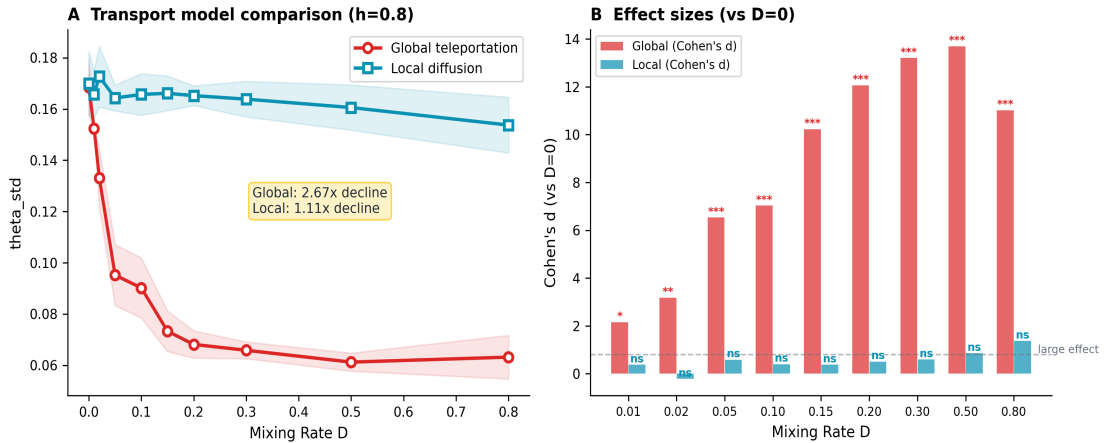


Figure 9. Transport model comparison at $h = 0.8$. (A) θ_{std} vs D under global teleportation (red) and local diffusion (cyan). Shaded bands: $\pm 1\sigma$ across seeds. (B) Cohen's d effect sizes (vs $D = 0$) for both transport models. Statistical significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

4.7 Colonization-limited regime

To test whether an intermediate dispersal optimum can emerge when some connectivity is mechanistically beneficial, we ran a colonization-limited sweep: local diffusion with abiogenesis reduced 5-fold (from 5×10^{-4} to 10^{-4} per cell per tick). In this regime, empty patches that suffer local extinction can only be rescued by neighbouring dispersers, creating the colonization-homogenisation tradeoff that underpins the intermediate dispersal optimum in metacommunity theory [9]. Figure 10 shows the results.

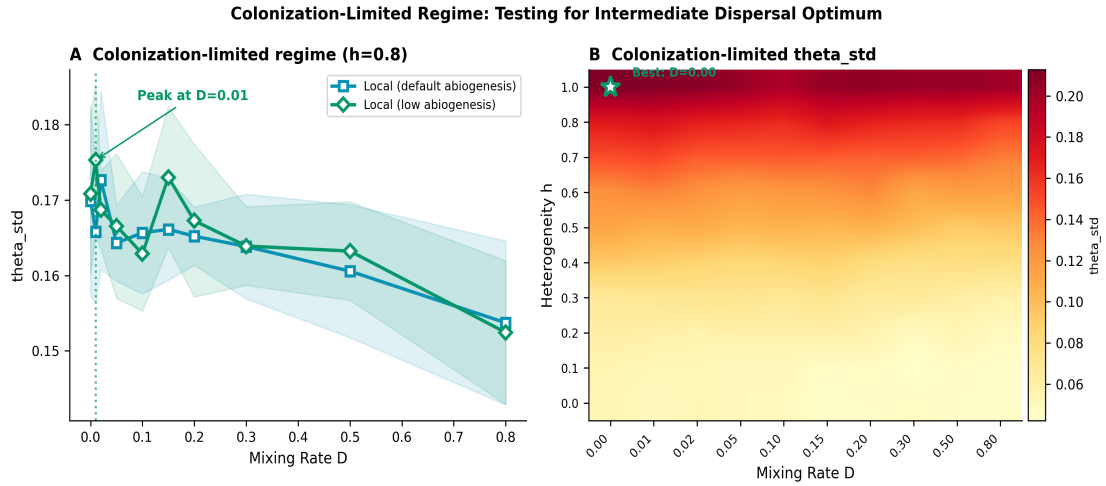


Figure 10. Colonization-limited regime. (A) θ_{std} vs D at $h = 0.8$ with default and reduced abiogenesis under local diffusion. (B) θ_{std} heatmap for the colonization-limited sweep. Star marks the parameter combination with highest diversity.

4.8 Statistical summary

Figure 11 presents the key result with full statistical annotations. Under global teleportation at $h = 0.8$, every non-zero D value shows significantly lower θ_{std} than $D = 0$ (all $p < 0.001$, Cohen's d ranging from ~ 1.5 at $D = 0.01$ to > 8 at $D = 0.5$). Under local diffusion, effect sizes are an order of magnitude smaller and most comparisons are not statistically significant at $\alpha = 0.05$ with 5 seeds. This confirms that the qualitative conclusion depends entirely on the transport model.

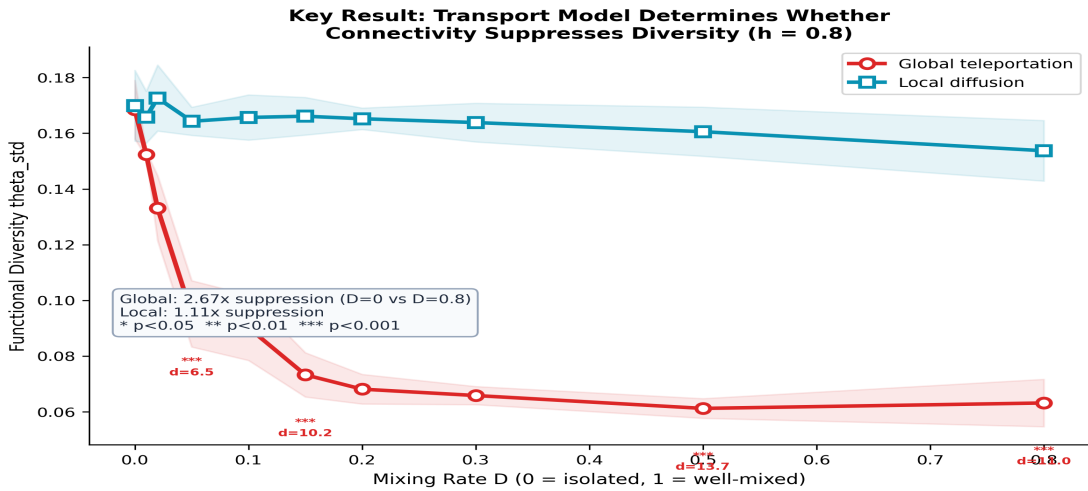


Figure 11. Key result with statistical tests. θ_{std} vs D at $h = 0.8$ for both transport models. Annotations show significance level (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$) and Cohen's d for each D vs $D = 0$ comparison.

5. Discussion

5.1 Transport model determines qualitative outcome

Our central finding is that the qualitative effect of connectivity on diversity depends critically on the transport mechanism. Under global teleportation, mixing suppresses diversity by $2.7\times$ ($D = 0$ vs $D = 0.8$ at $h = 0.8$; Cohen's $d > 5$, $p < 0.001$). Under local diffusion, the same comparison yields only $\sim 1.1\times$ suppression — statistically insignificant with 5 seeds. This mirrors the finding of Czárán et al. [12], who showed that lower diffusion rates monotonically favour replicator coexistence in their Metabolic Replicator Model, but that the effect becomes weak at realistic diffusion scales. Our contribution is to demonstrate this transport-dependence explicitly within a single model framework

using a full (D, h) phase diagram.

The mechanism is straightforward under teleportation: specialists are randomly relocated to distant, mismatched environments where their fitness drops sharply, driving competitive exclusion toward generalist strategies. Local diffusion avoids this because each step moves a replicator only one cell, so environmental mismatch is small and niche specialisation is preserved. This distinction matters for habitability predictions: real hydrospheres have diffusive and advective transport, not teleportation, so the strong suppression observed under global mixing is likely an overestimate of what would occur in nature.

5.2 Comparison with prior work

Our results are consistent with the prior literature on spatial prebiotic replicators. Szabó et al. [11] showed that limited dispersal on mineral surfaces promotes higher replicator efficiency through spatial reciprocity. Czárán et al. [12] is the most directly comparable study: they swept diffusion parameters in a metabolically coupled replicator system on a cellular automaton and found that coexistence of all replicator types requires sufficiently limited mobility. Their "critical diffusion rate" beyond which coexistence breaks down is analogous to our observation that global teleportation breaks niche specialisation. The key advance of this study is (i) the construction of an explicit (D, h) phase diagram with a niche-based fitness function, (ii) the systematic comparison of global vs local transport within a single framework, and (iii) the test of whether the intermediate dispersal optimum from metacommunity ecology [9, 10] has a prebiotic analogue.

5.3 Implications for habitability

For the broader question of planetary habitability, our results suggest a nuanced picture. Under any transport model, heterogeneous environments favour diversification (Figure 2). However, the strength of the isolation advantage depends on the transport mechanism. Under global mixing, the hierarchy is stark: shallow ponds outperform oceans by $\sim 3.5\times$. Under local diffusion, the difference is much smaller. This means that habitability predictions based on connectivity alone may be less reliable than those based on environmental heterogeneity.

Within the context of our group project on hydrospheres and life, this sub-project provides the ecological modelling component. Aims 1 and 2 determine which hydrosphere types are physically and chemically plausible; our model quantifies which of those types best supports the diversification step — a necessary precursor to the emergence of complex life.

5.4 Limitations

The model is deliberately abstract. Replicators are characterised by a single niche trait (θ) and lack explicit biochemistry (no RNA folding, no metabolic networks). The 40×40 grid is small compared to real-world environments; 80×80 tests (not shown) indicate $\sim 18\%$ lower θ_{std} , suggesting finite-size effects. Environmental patterns are static, whereas real hydrospheres exhibit dynamic cycling (tides, seasons, geological change). Abiogenesis contributes a neutral diversity floor of $\theta_{\text{std}} \approx 0.012$, which is not subtracted from reported values. Population density is weakly confounded with D ($\sim 4\%$ higher at $D = 0.8$ than at $D = 0$), but this effect is small compared to the measured diversity changes.

6. Conclusions

1. Under global teleportation, isolation maximises diversity. θ_{std} reaches its maximum at $D = 0.0$ for every heterogeneity slice, with a 62.5% decline at $h = 0.8$ ($p < 0.001$, Cohen's $d > 5$).

2. The result is transport-model dependent. Under physically realistic local diffusion, the suppression effect nearly disappears ($\sim 1.1\times$ vs $\sim 2.7\times$), consistent with Czárán et al. [12].

3. **Shallow pond networks are the strongest named scenario.** Shallow Ponds outperform Deep Ocean by 3.5× on functional diversity.

4. **The result is robust to selection strength.** Sensitivity analysis across $\beta \in \{10, 20, 40\}$ preserves the qualitative pattern under both transport models.

5. **The intermediate dispersal optimum from metacommunity ecology does not clearly emerge in this prebiotic model.** Even under colonization-limited conditions (low abiogenesis, local diffusion), the diversity maximum remains near the boundary $D \approx 0$.

6. **Environmental heterogeneity is the primary driver of diversity** under both transport models. The (D, h) phase diagram shows that h has a larger effect on θ_{std} than D under local diffusion.

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